GG-S-161D
May 1, 1985
SUPERSEDING
GG-S-161C
3 May 1977

FEDERAL SPECIFICATION

SCALES, DRAFTING, AND SCALES, PLOTTING

(GENERAL SPECIFICATION)

This specification was approved by the Assistant Administrator, Federal Supply and Services, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the general requirements for scales made of wood, plastic, metal, and combinations thereof.

1.2 Classification. Scales furnished under this specification shall be of the types, shapes, compositions, grades, and sizes as specified and identified by a type designation number (see 6.2).

1.2.1 Type designation number. Scales covered by this specification shall be identified by a type designation number formed as indicated below. This type designation number is intended for cataloging and ordering purposes and not for surface printing on the scales. Sub-paragraphs 1.2.1.1 through 1.2.1.10 indicate the options available for each type scale.

FSC 6675

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GG-S-161	X	X	X	X	X
Spec. No. (1.2.1.1)	Type (1.2.1.2)	Shape (1.2.1.3)	Composition (1.2.1.4)	Grade (1.2.1.5)	Size (1.2.1.6)

X	X	X	X
Scale graduation (1.2.1.7)	Numbering (1.2.1.8)	Style (1.2.1.9)	Sheath, if required (1.2.1.10)

1.2.1.1 The scale specification number shall be GG-S-161.

1.2.1.2 The type of scales shall be designated as follows:

Type I - Scales, drafting: architect's.
Type II - Scales, drafting: engineer's
Type III - Scales, drafting: mechanical.
Type IV - Scales, drafting: combination English and metric measurement system.
Type V - Scales, drafting: surveyor's.
Type VI - Scales, drafting: decimal-inch dimensioning.
Type VII - Scales, plotting.
Type VIII - Scales, plotting: metric.
Type IX - Scales, plotting: feet and yards.
Type X - Scales, plotting: meters and yards.
Type XI - Scales, plotting: soil survey.
Type XII - Scales, drafting: enlarging.
Type XIII - Scales, drafting: metric proportional.

1.2.1.3 The shape of the scales shall be designated as follows (see figure 1):

1 - Two bevels, flat, opposite (parallelogram).
2 - Two bevels, flat, same side (trapezoid).
3 - Alternate shape, two bevels, composition G only.
4 - Four bevels (flattened hexagon or lozenge).
5 - Single bevel, flat.
6 - Six bevels, triangular, flat or regular.
7 - Six bevels, triangular, facet or concave.

1.2.1.4 The composition of the scales shall be as follows:

A - Wood with plastic facings (includes B).
B - Wood with complete plastic coating except ends (excludes A).
C - Plastic.
D - Wood.
E - Aluminum alloy (plastic coated).
F - Aluminum alloy (anodized).
G - Steel.

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1.2.1.5 The grade of scale shall be designated as follows:

1 - High precision.
2 - Regular precision (to be supplied unless grade 1 is specified).

1.2.1.6 The size of the scales shall be designated as follows:

A - 10 inches.
B - 12 inches/30 centimeters.
C - 18 inches.
D - 50 centimeters.
E - 24 inches.
F - 6 inches/15 centimeters (pocket) except that certain metric scales may be 10 centimeters.

1.2.1.7 The scale graduations shall be designated as follows (see figure 1):

1 - Fully divided.
2 - Open divided.

1.2.1.8 The numbering of the scales shall be designated as follows (see figure 1):

A - Direct numbering.
B - Inverse numbering.
C - Double numbering.
D - Boustrophedon numbering.
E - Single numbering.

1.2.1.9 The style shall be either the English or metric measurement system displayed as decimal, fractional, or ratios of the various units of measurement within each system. Styles shall be as specified in each specification sheet and shall be designated in alphabetic sequence.

1.2.1.10 Sheaths shall be designated as follows (see 3.8):

1 - Leather.
2 - Artificial leather.
3 - Heavy vinyl plastic.
4 - Heavy paper or plastic.
5 - Plastic or cloth-covered cardboard or calendered cardboard.
6 - None required.

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2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on date of invitation for bids or solicitation for offers, form a part of this specification to the extent specified herein.

Federal Specifications:

L-P-349	- Plastic Molding and Extrusion Material, Cellulose Acetate Butyrate.
L-P-410	- Plastic, Polyamide (Nylon), Rigid: Rods, Tubes, Flats, Molded and Cast Parts.
L-P-535	- Plastic Sheet (Sheeting): Plastic Strip: Poly (Vinyl Chloride) and Poly (Vinyl Chloride-Vinyl Acetate), Rigid.
L-P-1183	- Plastic Molding Material, Acrylonitrile-Butadiene-Styrene (ABS), Rigid.
TT-E-529	- Enamel, Alkyd, Semigloss.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-665	- Boxes, Paperboard, Metal Edged and Components.
PPP-B-676	- Boxes, Setup.

(See supplement 1 for applicable specification sheets.)

Federal Standards:

FED-STD-123	- Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

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(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications:

MIL-P-116	- Preservation, Methods of.
MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-V-12276	- Varnish, Phenolic, Baking.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-C-22750	- Coating, Epoxy-Polyamide.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

Military Standards:

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-889	- Dissimilar Metals.

(Copies of Military Specifications and Standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Federal Regulations:

41 CFR 1-1.25	- Recovered Materials.

(The Code of Federal Regulations (CFG) is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 702	- Cast Methacrylate Plastic Sheets, Rods, Tubes, and Shapes.
ASTM D 707	- Cellulose Acetate Butyrate Molding and Extrusion Compounds.

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ASTM D 1781 - Climbing Drum Peel Test for Adhesives.
ASTM D 1788 - Rigid Acrylonitrile - Butadiene - Styrene (ABS) Plastics.
ASTM D 2016 - Moisture Content of Wood, Tests for.
ASTM D 2123 - Rigid Poly (Vinyl Chloride-Vinyl Acetate) Plastic Sheet.
ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. *REQUIREMENTS*

3.1 Description. Scales shall consist of linear strips of the specified materials with such graduations as are specified marked on the edges.

3.1.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets.

3.2 Material. *Material shall be as specified herein. Materials not* specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.2.1 Regulatory requirements. *The offeror/contractor is encouraged to use* recovered materials in accordance with Public Law 94-580 (41 CFR 1-1.25) to the maximum extent practical.

3.2.2 Material deterioration and control. *The scales shall be fabricated from* compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.2.2.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.2.2.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

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3.2.3 Wood. Wood shall be boxwood, pearwood, ramin wood, or selected hard maple free from knots, blemishes, or other defects, and shall be thoroughly seasoned to a moisture content of from 5 to 9 percent. Wood shall be straight grained and clear throughout, and free from imperfections such as chips, dents, or flaws.

3.2.4 Plastics.

3.2.4.1 Plastic sheets. Plastic sheets shall be one of the following:

a. Cellulose acetate butyrate conforming to ASTM D 707, grade MS or S2, or to L-P-349, grade MS or S2.

b. Methyl methacrylate conforming to ASTM D 702, type II, grade 2.

c. Nylon conforming to L-P-410.

d. Polyvinyl chloride conforming to L-P-535, or to ASTM D 2123, type I, matte finish.

e. Acrylonitrile-butadiene-styrene conforming to ASTM D 1788, type 1, grade 2, or to L-P-1183, type III or V.

3.2.4.2 Plastic coating. Plastic coatings used on metals shall be melamine urea formaldehyde conforming to TT-E-529, type II.

3.2.5 Steel. Steel shall have a minimum chromium content of 12 percent.

3.3 Storage temperature. Scales shall not be damaged by storage in an ambient temperature of 155° F for 4 hours or minus 65° F, without solar radiation, for 8 hours.

3.4 Shapes. Shapes shall be as shown in figure 1.

3.4.1 Shapes 1, 2, 4, and 5. Flat scales, except pocket scales, shall be 1-1/4 inches, plus or minus 1/8 inch, wide; pocket scales shall be 1 inch, plus or minus 1/16 inch, wide.

3.4.2 Shapes 6 and 7. Triangular scales shall be 1 inch, plus or minus 1/16 inch, on each side. Bowing shall not exceed 0.010 inch if it is upward in the center, nor exceed 0.005 inch if bowed downward at the center (ends upward) on two of the three sides, 0.003 inch if it is upward in the center, or 0.0015 inch if it is downward at the center on the third side.

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3.4.3 Shape 3. Shape 3 shall be used as an alternate for shape 1 or 2 when specified herein (see 1.2.1.3).

3.5 Graduations, numbering, and styles. Graduations, numbering, and styles shall be as specified herein and in the applicable detail specification sheets. Graduations and numbering shall be centered in the overall length of the scale; shall be clear and distinct; shall be fine and clean cut; shall cause no bulging or chipping of the remaining surface; and shall not smear or become indistinct when rubbed with a damp cloth. The filling shall be black, unless different systems of measurements, scale ratios (see 6.3.1), or other systems are on the same scale (on different edges), in which case one will be marked in black and the other in red, or one system shall be on a white facing and the other on a yellow-green facing.

3.5.1 Graduations. Graduations shall be fully divided, open divided or double divided (see 6.3.3, 6.3.4, and 6.3.5, respectively). Width of graduation marks shall be 0.003 inch, plus or minus 0.001 inch. Graduation lines shall be straight, uniform, parallel, and free from breaks or ragged edges. Graduation lines of decreasing lengths shall represent fractional aliquots (see 6.3.6) of each scale in descending order. Lines representing equal fractional aliquots shall be of equal length and readily distinguishable by their relative length from graduation lines representing other fractional aliquots. On fully-divided scales, the length of the shortest graduation lines shall be twice the smallest graduation spacing, plus 0.005 inch minus 0 inch, but shall not exceed 1/8 inch.

3.5.2 Numbering. Numbering shall be double, direct, boustrophedon, or single (see 6.3.7 through 6.3.10, respectively). Each significant graduation mark shall be numbered from zero at the index mark.

3.5.3 Styles. Scale divisions shall follow styles for the English or metric measuring system as specified in 1.2.1.9 and the specification sheet for each type scale as specified in 1.2.1.2.

3.6 Construction. The overall length of scales shall extend not less than 1/4 inch and not more than 1/2 inch beyond the graduations at each end, except that pocket scales shall extend not less than 3/16 inch but not more than 3/8 inch. Ends shall be square. The corners on all scales and the edges on ends of triangular scales shall be slightly rounded. Except for composition D scales, plotting edges shall be between 0.01 inch and 0.02 inch in width, with the variation along any edge not to exceed 0.01 inch. Scales shall not be damaged or warped by exposure to 90 percent relative humidity for 24 hours. Bowing shall not exceed 0.010 inch if it is upward in the center, or 0.005 inch if it is downward at center (ends upward). Scales shall show no warpage or twist when placed on a plane surface. Edges shall be straight to within 0.01 inch. Scales shall not shatter, split, separate, chip, crack, or nick when dropped 3 feet onto a hard surface.

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3.6.1 Plastic facings and coatings. Plastic facings and coatings shall be bonded to the base material in such a way that there shall be no tendency for the plastic to peel, loosen, or separate. Plastic facings on flat scales shall be inlaid flush with the rest of the surface. Facings shall accommodate all graduations and numbering. All colored coatings or facings shall be white or "eyesaver" yellow-green (see 6.3.11), except that the fully-divided scales having 50 or more graduations per inch shall be white only.

3.7 Finish. Scales shall be free from visible scratches or other imperfections and shall have a smooth satin or matte finish to minimize reflections.

3.8 Sheath. When specified (see 1.2.1.10), each scale shall be provided with an individual sheath. Sheaths for flat scales shall be open at one end; those for triangular scales shall be open at both ends and sufficiently close fitting to require a slight pressure to remove the scale. Unless otherwise specified herein, sheaths shall be leather treated to resist fungus, artificial leather, or heavy vinyl plastic; in addition, for military procurements only, sheaths for flat scales other than pocket scales may also be made of heavy paper or plastic and those for triangular scales may be made of heavy paper, plastic- or cloth-covered cardboard, or calendered cardboard.

3.9 Markings. Markings shall be legible and shall have the same permanency as the graduations. All markings shall be clean cut, shall cause no bulging or cupping of the surface, and shall not interfere in any way with the legibility of the graduations and numbering. Each scale shall be marked with the manufacturer's identification and with the words "Property U.S. Govt". The appropriate unit of measurement shall be marked near the zero of each set of graduation numbers. If all edges of a plotting scale are graduated in the same ratio, that ratio shall be marked near the center of the scale; if in different ratios, the appropriate ratio shall be marked near the zero of the graduation numbers. Pocket scale sheaths shall be marked "U.S.".

3.10 Compositions.

3.10.1 Compositions A, B, C, and D. Flat scales in compositons A, B, C, and D shall be 3/16 inch, plus or minus 1/32 inch, thick.

3.10.1.1 Composition B. Composition B scales, except for the ends, shall be completely covered with plastic.

3.10.1.2 Composition D. Composition D scales shall have not less than two coats of clear hard-drying varnish conforming to MIL-C-22750, type I, which shows the grain of the wood.

3.10.2 Compositions E, F, and G. Flat metal scales shall be 1/8 inch, plus or minus 1/32 inch, thick. Markings and graduations shall be protected by not less than two coats of clear varnish conforming to MIL-V-12276, type III, or, on composition F, by anodizing in accordance with MIL-A-8625, type I, to a thickness of 0.0001 inch.

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3.11 Accuracy.

3.11.1 Grade 1. Grade 1 scales shall be engine divided only. The error in distance between any two adjacent graduations shall be less than 0.002 inch at 70° F, plus or minus 5° F, and 50 percent relative humidity. The error in the graduated overall length shall not exceed 0.001 inch, plus 0.001 inch, for each 4 inches of overall length and for any additional fraction of such graduated length (see 6.3.12).

3.11.2 Grade 2. The error in distance between any two adjacent graduations shall be less than 0.003 inch at 70° F, plus or minus 5° F, and 50 percent relative humidity. The accumulative error shall be equally distributed throughout the graduated lengths of the scales and shall not exceed the following dimensions and tolerances:

Length of graduated area	Tolerance, plus or minus
(inches)	(inches)
6 or less	0.004
10	0.006
12	0.008
18	0.010
20 or greater	0.012

3.12 Workmanship. Scales shall be free from defects that could impair their accuracy or serviceability. The scales shall have no rough edges, burrs, cracks, chips, or scratches.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.3).

b. Inspection of preparation for delivery (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling.

4.3.1.1 For examination. Sampling for examination shall be in accordance with MIL-STD-105, inspection level III.

4.3.1.2 For tests. Sampling for tests shall be in accordance with MIL-STD-105, inspection level S-2, for tests specified in 4.4.2.2, 4.4.2.3, and 4.4.2.8, and inspection level S-4 for all other tests.

4.3.2 Examination. Samples selected in accordance with 4.3.1.1 shall be examined as specified in 4.4.1. For grade 1 scales, AQL shall be 1.0 percent defective for major defects and 1.5 percent defective for minor defects; for grade 2 scales, AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.3.3 Tests. Samples selected in accordance with 4.3.1.2 shall be tested as specified in 4.4.2 in the order listed, except where otherwise indicated. AQL shall be 1.5 percent defective for grade 1 scales and 4.0 percent defective for grade 2 scales.

4.4 Inspection procedure.

4.4.1 Examination. The scales shall be examined as specified herein for the following defects:

Major

101. Classification not as specified.
102. Materials not as specified.
103. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
105. Contractor does not have documentation available for identification of material, material finishes or treatment.
106. Graduations and numbers not as specified.
107. Construction not as specified.
108. Plastic not properly bonded to base material, where applicable.

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109. Dimensions not as specified.
110. Sheath missing, when applicable.
111. Workmanship not as specified.

Minor

201. Colors not as specified.
202. Finish not as specified.
203. Sheath not as specified, when applicable.
204. Markings missing or not as specified.

4.4.2 Tests.

4.4.2.1 Moisture content. A sample of each lot of the wood used in composition A or D scales shall be tested by either the oven drying method or the electrical resistance method in accordance with ASTM D 2016. A moisture content less than 5 percent or more than 9 percent shall constitute failure of this test.

4.4.2.2 Environmental tests. The following tests are applicable only to scales having plastic components or coverings.

4.4.2.2.1 Storage temperature. The scales shall be placed in an ambient temperature of not less than 155° F and allowed to remain at this temperature for not less than 4 hours. The scales shall then be removed to normal room temperature and examined after they have stabilized. If there is no evidence of damage, the scales shall be placed in an ambient temperature of not more than minus 65° F without solar radiation for not less than 8 hours. The scales shall then be removed to normal room temperature and examined after they have stabilized. Any evidence of warping, peeling, cracking, discoloration, separation, or any other damage shall constitute failure of this test.

4.4.2.2.2 Warp, bow, twist, and linear expansion. Sample scales shall be subjected to a relative humidity of not less than 90 percent at 120° F for not less than 24 hours or, as an alternative, may be placed in water having a temperature of not less than 120° F for not less than 30 minutes. Immediately following one of these exposures the scale shall be tested in accordance with 4.4.2.7, 4.4.2.6, and 4.4.2.5, in that order, as expenditiously as possible (these tests need not be repeated on scales having plastic components or coverings). Failure to meet the failure criteria of any of these tests shall constitute failure of this test.

4.4.2.3 Drop test. The scale shall be dropped from a height of 3 feet onto a hard surface, landing once on each end and once on each side. Any evidence of splitting, cracking, chipping, separation, or other damage shall constitute failure of this test.

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4.4.2.4 Markings permanency. A clean white cloth shall be moistened with water and rubbed, under pressure, along the full length of one of the graduated surfaces of the scale, keeping the same portion of the cloth in contact with the scale at all times. The graduations and markings shall be rubbed not less than 10 times in each direction. This test shall be repeated on each color. Any adverse effect on the graduations or markings, or any traces or coloring matter on the cloth, shall constitute failure of this test.

4.4.2.5 Straightness. Straightness of scales shall be tested by one of the following methods.

4.4.2.5.1 Flat method. Each edge of the scale shall be placed against a machinist's flat or true straightedge, and any deviation from this reference surface shall be measured with a thickness gage. Evidence of edges not straight to within 0.010 inch shall constitute failure of this test.

4.4.2.5.2 Drafting method. A good grade of drawing paper shall be fastened to a drawing board. Two pins shall be driven vertically 1/4 inch into the board, spaced approximately 1/2 inch less than the length of the scale to be tested. The edge of the scale being tested shall be placed against the pins and a line shall be drawn along the edge with a sharp, 3H or harder, lead pencil. The scale shall be rotated 180 degrees, the same edge placed against the pins, and a second line drawn. The face of the scale shall be held in contact with the drawing paper when testing flat scales in reverse. The distance between the two lines shall be measured at intervals of less than 2 inches with a shop microscope of not less than 40X or by any other method of at least equal precision. Any variation in distance between the two lines greater than 0.010 inch shall constitute failure of this test.

4.4.2.6 Planeness. Scales shall be tested for bow and twist by laying the scale on a machinist's flat so that both ends of the scale are on the flat, and the clearance shall be checked with a thickness gage. Clearance at the center more than 0.010 inch or more than 0.005 inch at the end shall constitute failure of this test.

4.4.2.7 Accuracy. By use of dividers and beam compass or by use of an optical comparator that will measure to plus or minus 0.001 inch, the spacing of the graduation marks shall be tested by comparison with the appropriate graduations of a test bar measured as accurate to 0.0004 inch by reference to a suitable length scale calibrated by the National Bureau of Standards. Tests shall be made at 70° F, plus or minus 5° F. Adjacent graduations shall be tested at sample intervals not more than 2 inches throughout the scale. The overall dimensions of the graduation on each edge or bevel shall be tested. Nonconformance to 3.5 and 3.11 shall constitute failure of this test.

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4.4.2.8 Strength of adhesive. Scales with facings not integral with the body of the scale, except those with both contact surfaces corrugated to improve the adhesion, shall be tested in accordance with ASTM D 1781, modified to the actual width of the facing, except, at the manufacturer's option, the conditioning procedures specified in 4.4.2.2 may be considered as the required conditioning. The test will follow any other test. A peel strength less than the equivalent of 15 pounds per inch of width shall constitute failure of this test.

4.5 Inspection of preparation for delivery.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following major defects. The AQL shall be 2.5 percent defective and each incorrect material, method, or container shall constitute one defect.

No.	Defect	A	B	Comm.
112.	Scales not preserved as specified.	5.1.1	5.1.2.1 5.1.2.2.1	5.1.3
113.	Preserved scales of like description not consolidated as specified.	5.1.1.1	5.1.2.1 5.1.2.2	
114.	Consolidation containers not as specified.	5.1.1.1	5.1.2.1 5.1.2.2	
115.	Box closure not as specified.	5.1.1.1	5.1.2.2.1 5.1.2.2.2	
116.	Scales of like description not packed as specified.	5.2.1	5.2.2.1 5.2.2.2	5.2.3
117.	Exterior containers not as specified.	5.2.1	5.2.2.1 5.2.2.2	5.2.3
118.	Closure of exterior containers not as specified.	5.2.1	5.2.2.1 5.2.2.2	5.2.3
119.	Gross weight exceeds specified maximum.	5.2.1	5.2.2.1	
120.	Strapping of exterior containers not as specified.	5.2.1	5.2.2.1	
121.	Alternate exterior container, when utilized, not as specified.		5.2.2.1	
122.	Alternate exterior container, when utilized, not reinforced as specified.		5.2.2.1	
123.	Marking not as specified.		5.3	

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5. PREPARATION FOR DELIVERY

5.1 Preservation. Preservation shall be level A, level B, or commercial, as specified (see 6.2).

5.1.1 Level A. Each scale, with or without sheath, shall be preserved in accordance with MIL-P-116, method IC-1. When the scales are furnished in a manufacturer's carton, they shall be preserved in accordance with MIL-P-116, method IC-2, using the manufacturer's carton as the container.

5.1.1.1 Consolidation. Scales of like description preserved as specified in 5.1.1, shall be consolidated together in quantities not to exceed 20 pounds in close-fitting boxes conforming to PPP-B-636, type CF, class weather-resistant, variety SW, grade W6c, style optional. Box closure shall be in accordance with method IV of the appendix thereto.

5.1.2 Level B.

5.1.2.1 Military procurement. Each scale, with or without sheath, shall be preserved as specified in 5.1 for level A, except that the boxes for consolidation shall conform to PPP-B-636, type CF, class domestic, variety SW, grade 125.

5.1.2.2 Civil agency procurements.

5.1.2.2.1 Preservation. Six or twelve scales of like description, with or without sheaths, shall be placed in a close-fitting box conforming to PPP-B-566, PPP-B-665, or PPP-B-676. The applicable variety, style, type and class of the box shall be optional. Scales without sheaths shall be individually wrapped with non abrasive material before placement in the box, to prevent damage. Box closure shall be in accordance with the applicable box specification.

5.1.2.2.2 Consolidation. Six boxes of scales of one stock number, preserved as specified in 5.1.2.2.1, shall be consolidated into a close-fitting box conforming to PPP-B-566, PPP-B-665, PPP-B-676 or PPP-B-636. The applicable variety, style, type and class of the box shall be optional except that boxes conforming to PPP-B-636, shall be class domestic, variety SW, type CF and grade 125. Box closure shall be in accordance with the applicable box specification.

5.1.3 Commercial. The scales shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

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5.2.1 Level A. Scales of like description, preserved as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type, style I, or PPP-B-621, class 2, style optional. The gross weight of the box shall not exceed 200 pounds. Box closure and strapping shall be as specified in the applicable specification or appendix thereto except that strapping shall be flat and the finish B.

5.2.2 Level B.

5.2.2.1 Military procurements. Scales of like description, preserved as specified in 5.1, shall be packed as specified in 5.2 for level A except that the boxes shall be domestic type or class. As an alternate, scales of like description, preserved as specified in 5.1, may be packed in close-fitting fiberboard boxes conforming to PPP-B-636, type CF, class weather-resistant, variety SW, grade V3c, style optional. The box closure shall be in accordance with method V of the appendix to the fiberboard box specification, the reinforcing of the sealed box with bands of reinforced tape shall be in accordance with the appendix to the fiberboard box specification and the gross weight shall not exceed the weight limitation of the fiberboard box specification.

5.2.2.2 Civil agency procurements. Scales of like description, preserved as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-636, type CF, class domestic, variety SW, grade 275, style optional. The gross weight shall not exceed the weight limitation specified therein and box closure shall be in accordance with method IV of the appendix thereto.

5.2.3 Commercial. Scales of like description, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Military procurement. In addition to any special marking required by the contract or order (see 6.2), all containers shall be marked in accordance with MIL-STD-129.

5.3.2 Civil agency procurements. Civil agent marking shall be in accordance with FED-STD-123.

5.3.3 Commercial. Commercial marking shall be in accordance with ASTM D 3951. Additionally, each shipping container shall be marked to show the gross weight and cube.

6. NOTES

6.1 Intended use. Drafting and plotting scales are for measuring and laying out linear dimensions. They are not intended for use as a substitute for straightedges, triangles, or T-squares. They are not intended for adjusting dividers or

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compasses on the scale; special engraved calibrating scales are available for this purpose. They are not intended for use in drafting machines, such scales being covered in MIL-D-3898. However, should it be necessary to use any of the shape 2 or 5 scales of this specification in drafting machines, this may be done by attaching chuck plates conforming to MS16885. Compositions A and C are generally intended for use indoors, and composition D for training and rough work.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number, and date of this specification.

b. Type designation number (see 1.2.1).

c. Degree of preservation and degree of packing required (see 5.1 and 5.2).

d. Any special marking requirements (see 5.3.1).

6.3 Definitions. The following definitions shall apply throughout this specification.

6.3.1 Scale ratio. Scale ratio equals representative fraction.

6.3.2 Face reading. An edge numbered to be read when nearest the user and above what is being measured (rather than read over the scale when the scale is below what is being measured) (see figure 1).

6.3.3 Fully divided. Each principal unit is fully subdivided throughout the scale, and the index graduation is at one end of the graduated length (see figure 1).

6.3.4 Open divided. Only the principal units are graduated; one or more units on the opposite side of the index graduation from the main portion of the scale are fully subdivided in the opposite direction (see figure 1).

6.3.5 Double divided. A direct open-divided scale superimposed on an inverse open-divided scale to coincide at as many points as possible (see figure 1).

6.3.6 Fractional aliquot. A fraction which, when divided into unity, leaves no remainder.

6.3.7 Double numbering. Two numbering series on one set of graduations. One series must be an exact multiple of the other (see figure 1).

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6.3.8 Direct numbering. A numbering series increasing in magnitude to the right (see figure 1).

6.3.9 Boustrophedon numbering. A double numbering, one series of which is direct, the other inverse (see figure 1).

6.3.10 Single numbering. Only one numbering series per set of graduations. May be in either direction.

6.3.11 Yellow-green. A color falling between color chips 23814 and 24552 of FED-STD-595 (this is approximately 5600 angstrom units).

6.3.12 Graduated length. The linear area actually occupied by graduations.

6.3.13 Drafting scales or edges. Scales or edges graduated in common units of linear measurement (without reduction or enlargement) subdivided into fractional or decimal parts of the unit.

6.3.14 Index graduation. The zero graduation from which distances are measured.

6.3.15 Inverse numbering. A numbering series increasing in magnitude to the left; the reverse of direct numbering (see figure 1).

6.3.16 Nominal length. A rounded dimension approximately equal to the graduated length.

6.3.17 Overall length. The total length of the scale blank equals graduated length plus end spaces.

6.3.18 Plotting scales or edges. Scales or edges graduated in units applicable only to the plotting of distances represented by a map ratio (a plotting scale may have some drafting edges on it).

6.3.19 Scale. A line standard as opposed to a rule, which is an end standard.

6.4 Plotting scales. The following list of map ratios lists those in standard use among Government agencies. Insofar as possible, procurement of plotting scales should be limited to these ratios:

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Fractional scale	Mi/in	Km/mm	In/mi	Mm/km	In/ 1000 ft	Ft/in	Agency
1:1,000,000	15.783	1.0000	0.06	1.00	0.01	83,333.33	Military AF C&GS (A)
1:500,000	7.891	0.5000	0.13	2.00	0.02	41,666.67	Military AF C&GS
1:250,000	3.946	0.2500	0.25	4.00	0.05	20,833.33	Military C&GS (A)
1:100,000	1.578	0.1000	0.63	10.00	0.12	8,333.33	Military C&GS (N)
1:62,500	0.986	0.0625	1.01	16.00	0.19	5,208.33	GS
1:50,000	0.789	0.0500	1.27	20.00	0.24	4,166.67	Military C&GS (N)
1:31,680	0.500	0.0317	2.00	31.57	0.38	2,640.00	GS
1:25,000	0.395	0.025	2.53	40.00	0.48	2,083.33	Military C&GS (N)
1:24,000	0.379	0.024	2.64	41.67	0.50	2,000.00	GS
1:15,840	0.250	0.0158	4.00	63.13	0.76	1,320.00	GS
1:12,500	0.197	0.0125	5.07	80.00	0.96	1,041.67	Military
1:12,000	0.189	0.0120	5.28	83.33	1.00	1,000.00	GS
1:7,920	0.125	0.0079	8.00	126.26	1.52	660.00	D/Agri
1:3,960	0.0625	0.0040	16.00	252.52	3.03	330.00	D/Agri

6.5 *Classification changes*. Changes in classification of the scales between this revision of the specification and the previous editions are as follows:

Old Designation	New Designation
Type I - Architect's-fractions of an inch to the foot.	Type I - Scales, drafting: architect's, fractions of an inch to the foot.
Type II - Engineers-decimal parts of an inch to the inch ("chain scales").	Type II - Scales, drafting: engineer's, decimal parts of an inch/ centimeter.
Type III - Mechanical-fractions of an inch to the inch ("size scales").	Type III - Scales, drafting: mechanical, fractions of an inch/meter ("size scales").
Type IV - Metric.	Type IV - Scales, drafting: combination English and metric measurement system.

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Old Designation	New Designation
Type V - Surveyor's decimal part of a foot to the foot.	Type V - Scales, drafting: surveyor's, decimal parts of a foot to the foot.
Type VI - Decimal-inch dimensioning.	Type VI - Scales, drafting: decimal-inch dimensioning.
Type VII - None.	Type VII - Scales, plotting.
Type VIII - None.	Type VIII - Scales, plotting: metric.
Type IX - None.	Type IX - Scales, plotting: feet and yards.
Type X - None.	Type X - Scales, plotting: meters and yards.
Type XI - None.	Type XI - Scales, plotting: soil survey.
Type XII - None.	Type XII - Scales, drafting: enlarging.
Type XIII - None.	Type XIII - Scales, drafting: metric proportional.

MILITARY INTERESTS:

Custodians

Army - ME
Navy - YD
Air Force - 99

Review activities

Army - GL
Navy - MC
DLA - GS

User activity

Air Force - 82

CIVIL AGENCY COORDINATING ACTICITIES:

FSS
NBS
OSS
SCS

PREPARING ACTIVITY:

Army - ME

Project 6675-0780

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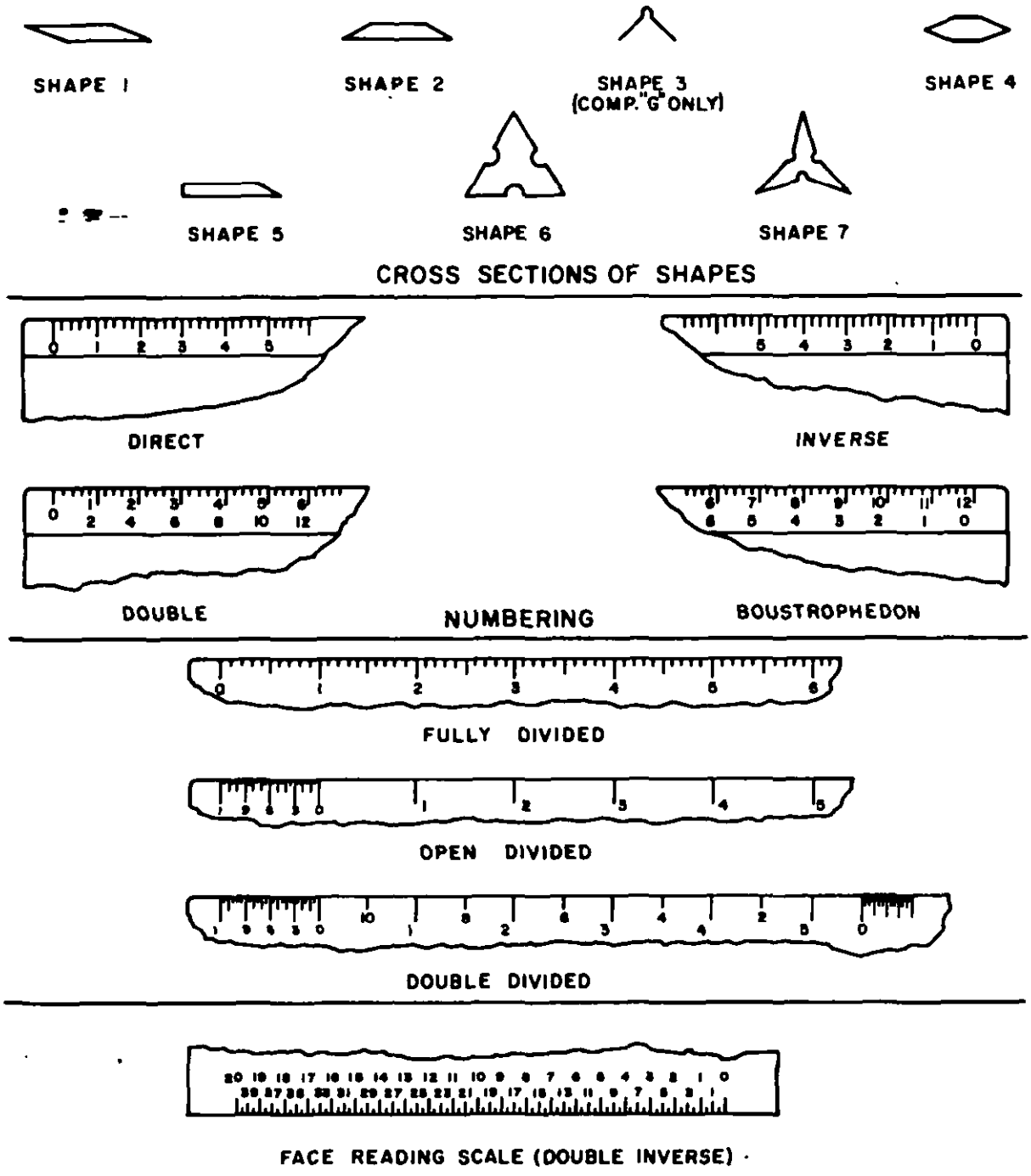


FIGURE 1. Scales, drafting.

X-3050B

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
GG-S-161D	Scales, Drafting, and Scales, Plotting

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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